Reframing Transport Poverty through a Community-Driven Design Approach

Oliver Weberg
Halmstad University
oliver.weberg@hh.se

Jesper Lund
Halmstad University
jesper.lund@hh.se

Vaike Fors
Halmstad University
vaike.fors@hh.se

Abstract

This study explores how a community-driven design approach can address human flourishing in the context of mobility. Drawing on design ethnography and co-design in two Swedish urban areas, the study surfaces alternative design ideas rooted in everyday life and community values. The research contributes to design for social sustainability by illustrating how place-based, participatory processes can make room for local narratives and values, offering a pathway for embedding wellbeing into the design process.

Keywords: Community-Driven Design; Human Flourishing; Social Sustainability; Design Ethnography

1. Introduction

The topic of how IS research can support human flourishing is currently debated. While Ciriello (2025) discusses how artefacts, exemplified through social media platforms, may influence humans' flourishing through the lens of Schopenhauer, this paper explores a community-driven design approach from a similar perspective. In the same manner as Ciriello (2025), we too have concerns regarding paradoxical functionalities depending on situational experiences and varying contexts, which we exemplify through the context of mobility. Digital mobility solutions, such as ride-hailing platforms, e-scooters, and Mobility-as-a-Service (MaaS), have been heralded as innovations capable of transforming urban transport (Miskolczi et al., 2021). However, emerging scholarship suggests that such technologies often reinforce, rather than redress, existing social and spatial inequalities (Sheller, 2018; Martens, 2017). These services, designed primarily for urban, digitally connected, middle-class users, risk marginalising vulnerable populations and undermining mobility justice (Dillahunt & Veinot, 2018). The digital divide, algorithmic discrimination, and profit-oriented service models can leave low-income, differently abled, migrant, or elderly populations with little or no access to mobility options, further deepening so-called transport poverty (Dorantes & Murauskaite, 2022). This article contends that an urgent need exists to examine alternative design approaches for digital mobility solutions that centre equity, inclusivity, and access to foster genuinely transformative and just mobility futures.

Addressing such complex social challenges calls for a holistic conceptual operationalisation that recognises the intertwined nature of economic, social, and ecological dimensions of sustainability, and that guides design efforts toward promoting human wellbeing. The Design for Social Sustainability (D4SS) paradigm offers such an operationalisation; it is defined as "design that advances the human wellbeing and flourishing of societies now and in the future" (Corsini & Moultrie, 2021). This approach emphasises integrating top-down and bottom-up strategies while being deeply attentive to local contexts, fostering participation, equity, and long-term community resilience (Boyer et al., 2016).

Inspired by the D4SS conceptualisation, this study presents findings from a design ethnographic project containing an area, Biskopsgården, officially labelled as "challenged" by the Swedish police due to crime-related activities and a culture of silence, as mentioned in Weberg et al. (2025), as well as another area, Forsåker, envisioned but not yet built. The study explored how design can contribute to social sustainability through research activities guided by the question: How can a community-driven design approach address human flourishing in mobility? Despite being part of Sweden's second-largest city, the areas suffer from transport poverty - defined here as the absence of accessible mobility services such as carpooling schemes, e-bikes, and shared transport options. Furthermore, relating to the grander definition through Martens (2017) and Sheller (2018), Biskopsgården is geographically disconnected from the other side of the city where many employed inhabitants work, involving spatial, temporal, financial, and social dimensions of exclusion.

Adopting a place-based, community-centric approach (Boyer et al., 2016; Corsini & Moultrie, 2021;

URI: https://hdl.handle.net/10125/112118
978-0-9981331-9-5
(CC BY-NC-ND 4.0)

Rossi, 2025), the project employed co-creative methods to identify and work with local values and practices. The study draws upon methods suitable for capturing local peculiarities and addressing the complexity of human flourishing. Drawing on design as a staging intervention (Mazé et al., 2011), design ideas were developed collaboratively with project actors to formulate alternatives, align visions, and test situated responses. The study contributes contextual insights to the discourse on design for social sustainability (D4SS), with particular emphasis on participation, recognition, and the development of resilient social structures (Eizenberg & Jabareen, 2017; Rossi, 2025) that can assist human flourishing (Ciriello, 2025).

The paper is structured as follows. First, we present a background section about designing for human flourishing and mobility justice, followed by a method section and an empirical section. Finally, we discuss the potential of a community-driven design approach and conclude the paper by answering the research question.

2. Designing for human flourishing

Design approaches that explicitly engage with social sustainability and human flourishing are gaining attention in response to complex socio-ecological challenges. One example is Design for Social Sustainability (D4SS), which focuses on advancing human wellbeing and supporting societies to thrive now and in the future (Corsini & Moultrie, 2021). D4SS aligns with growing recognition that flourishing is not only about individual happiness but also a collective and social phenomenon (Jones, 2017; Nakata, 2025). Human wellbeing is deeply embedded in the communities and cultures people inhabit, suggesting that designers must create conditions for these social and cultural ecosystems to flourish (Jones, 2017). This understanding positions flourishing and social sustainability as mutually reinforcing concepts, much like environmental sustainability is linked to the health of natural ecosystems. Some scholars even argue that building a flourishing society should be the aim of sustainability efforts (Jones, 2017; Corsini & Moultrie, 2021; Nakata, 2025). Design practices such as participatory-, transition-, systemic-, and D4SS all work toward these goals by engaging directly with communities. Rather than offering isolated or prescriptive solutions, these approaches support co-creation with local stakeholders to generate alternatives aligned with shared values and long-term aspirations (Corsini & Moultrie, 2021). However, translating these ideals into practice is difficult. Key concerns include defining success, measuring impact, and avoiding the reproduction of existing inequalities (Eizenberg & Jabareen, 2017), as well as designing locally suitable methods (Corsini & Moultrie, 2021).

Jones (2017) proposes the development of frameworks for flourishing cultures - design and policy models that embed wellbeing into innovation and systems change. These models aim to reimagine design as a vehicle for sustaining culture and community life, expanding its role beyond problem-solving toward actively fostering human flourishing. Nakata (2025) contributes to this perspective by proposing a two-path research agenda that connects design, innovation, and wellbeing. One path focuses on reshaping organisational structures to support societal goals better, while the other explores how design processes can generate innovations that directly improve individual and collective wellbeing. In this context, flourishing is understood not merely as fulfilling material needs but as an expansive form of wellbeing encompassing life satisfaction, personal growth, and a sense of meaning and purpose. Pohlmeyer (2017) complements this view by emphasising how design could facilitate short-term emotional uplift and longer-term psychological enrichment. The author argues that artefacts and systems can be intentionally designed to support users' pursuit of meaningful goals, relationships, and personal virtues. For instance, a product might be enjoyable to use and encourage self-reflection, connection with others, or the development of one's strengths.

Together, Nakata's structural agenda and Pohlmeyer's experiential focus illustrate how design for human flourishing can operate on multiple levels - strategic, systemic, personal, and emotional. This broadened view shifts the definition of successful design from purely functional or economic outcomes towards more profound contributions to human fulfilment. It suggests that design can become a possibility-driven discipline that cultivates creativity, empathy, and wellbeing by shaping environments and experiences that enable people and communities to thrive. Because flourishing spans multiple fields, it cannot be pursued in disciplinary isolation. Embedding it within design and innovation processes helps ensure that human and community wellbeing remain central concerns. Designing for meaningful engagement can significantly reinforce social sustainability. Communities with rich cultural life and strong social ties are better able to provide the resilience needed for people to flourish, particularly in times of crisis (Jones, 2017; Eizenberg & Jabareen, 2017).

Furthermore, Ciriello (2025), drawing on Schopenhauer, questions the assumption of a pure rational free will and instead calls for design approaches that respect human dignity and deeper impulses that reach beyond basic needs. When these impulses are situated in relation to the intentions and interpretations

of different stakeholders, they reveal the complexity of how artefacts acquire meaning within particular cultural and social contexts. In this sense, Ciriello's perspective emphasises social sustainability by highlighting the importance of subjective experiences in sustaining cultural life and resilient communities. Designing with an awareness of such dynamics fosters compassionate interactions that avoid imposing unrealistic standards, thereby enabling individuals and communities to flourish through meaningful and sustainable engagement.

To address these issues, that is, the tension between subjective experience, stakeholder interpretation, and the cultivation of social sustainability, scholars advocate for place-based, process-oriented design supported by qualitative methods such as ethnography, participatory action research, and semi-structured interviews (Boyer et al., 2016; Corsini & Moultrie, 2021). These approaches help uncover local values and enable inclusive, context-sensitive solutions. A design culture grounded in holistic thinking, adaptability, and resilience is better positioned to support long-term human flourishing (Rossi, 2025). Rather than prescribing fixed solutions, design research can stage "microcosms" that make societal challenges tangible, fostering critical reflection, social learning, and collaborative action (Mazé et al., 2011). In this context, design ethnographic approaches have proven to be powerful means of exploring both the material and social dimensions of lived experience, enabling participatory co-learning processes that centre local knowledge and relationships (Baskerville & Meyers, 2015; Pink et al., 2022; Smith et al., 2024; Fors et al., 2025; Weberg et al., 2025).

Yet, significant gaps remain - particularly in understanding how to navigate complex trade-offs and measure the broader impacts of innovation. Engaging with these challenges is essential for advancing both design theory and the real-world wellbeing of individuals and communities. As Nakata (2025) argues, because human flourishing spans multiple domains, real progress requires collaboration across disciplines and a commitment to embedding wellbeing at the core of design and innovation processes (Corsini & Moultrie, 2021). Ciriello (2025) extends the need to investigate further motivations for engagement and how this engagement can foster empathetic support that includes diverse experiences, which may alleviate suffering. Responding to this call, our study explores how a design ethnographic, community-driven design approach can serve as a practical means of addressing human flourishing. By focusing on local engagement, co-creation, and lived experience, this approach offers a grounded pathway for translating the concept of flourishing from theory into situated, meaningful design practice.

3. Method

This article examines how stakeholders in a residential area engaged with ethnographic insights generated through fieldwork, focusing not on the ethnography itself, as detailed in Weberg et al. (2025), but rather on the subsequent processes through which these insights were mobilised. Specifically, the study explores how we co-designed and facilitated workshops with citizens (actors), municipal actors, planners, and other relevant stakeholders, using our ethnographic material as a shared foundation for developing context-sensitive mobility solutions. The article thus sheds light on the interpretive and practical work involved in translating situated, lived experience into actionable design. Following Boyer et al.'s (2016) process-oriented, place-based framework, we did this through a design ethnographic approach (Pink et al., 2022; Fors et al., 2022; Smith et al., 2024), grounded in conventional ethnographic methods, to explore and develop a community-driven design to address mobility issues. Through this approach, we co-created not only design outcomes but also the activities and processes themselves with the ambition to uncover systemic challenges and explore the so-called "wicked problems" embedded in local mobility systems (Mazé et al., 2011; Ebbeson & Fors, 2023), while identifying community values that could inform future design. Drawing on traditions from Participatory Design and critical futures, we also questioned dominant narratives about the area by staging interventions that opened space for dialogue and collective imagination (Fors et al, 2025). Through fieldnotes, observations, and reflective analysis of what happened before, during, and after the stakeholder workshops - guided by O'Reilly's (2012) principles for ethnographic methods - we investigated how mobility could be reimagined through locally embedded, community-driven design together with stakeholders in the area. Such reflective analysis allowed us to explore what was designed and how and why it emerged from the specific social, cultural, and spatial context.

In this project, we began with a shared activity involving key stakeholders to establish a common ground according to design ethnographic principles and practices (Ebbesson et al., 2024; Fors et al., 2025) before moving into citizen engagement. The work with citizen engagement, method development, and design ideas included seven activities: five workshops and two participant observations. All workshops, observations, and co-design activities were carried out by the core research team, consisting of three researchers with backgrounds in participatory design, design

ethnography, and social sustainability. Researchers shared roles across planning, facilitation, note-taking, synthesis, and follow-up. Community champions were involved as facilitators and co-hosts in activities to support community translation and trust-building.

Project stakeholders were invited through project partner networks and local governance structures. At the same time, researchers and community champions recruited residents in collaboration with civil society organisations engaged in citizen dialogue. No formal demographic surveys were conducted among residents due to a noted and documented scepticism toward research teams. Instead, ethical engagement relied on a simple contract and informal introductions during and through participation. This ensured inclusive access, especially in Biskopsgården, where many participants had prior experiences of exclusion from planning processes.

The first three-hour workshop included fourteen project stakeholders, ranging from professionals with expertise in construction, parking, housing, and transport. The workshop aimed at introducing the problem area related to future mobility hubs with a presentation on state-of-the-art research and phenomena, where the participants worked in two groups to collaboratively communicate a common ground through a canvas created by us, focusing on what a mobility hub could be.

The second workshop was held in a local residential setting in Biskopsgården and aimed to explore current mobility routines, where 26 residents participated in a three-hour session. Considering initial concerns around trust and institutional affiliations, we deliberately did not collect standard demographic data. Instead, participant characteristics were understood through dialogue during the session. Most participants were foreign-born and ranged in age from approximately 13 to 60. The group included mothers with young children, older individuals not currently employed, and younger participants in education at different levels. Only one person reported access to a private car shared with a relative, while the rest of the group were without either a car or a driving license. At least half of the participants expressed that public transport was unaffordable, particularly the monthly bus and tram cost. Recruitment was supported by the local civil society organisation dedicated to promoting citizen dialogue.

The third workshop was conducted online with two intending buyers of apartments in Forsåker, aimed at exploring and understanding current routines and what would be the dream and nightmare scenarios of a potential car-free future. Following the three workshops, two participatory observations were carried out. One focused on a community champion in Biskopsgården. One of the researchers followed the champion for a day to understand local values using a contextual inquiry technique. The second participatory observation focused on a future resident of Forsåker. This participant was in the transition of moving away from the privately owned car. This person's experiences and explanations assisted us in understanding the barriers and enablers of adopting mobility services as a substitute for the car. One of the researchers moved around with the participant, following his everyday activities during the day, primarily focusing on how non-car-centric transport solutions were used. After analysing gathered material and designing two visions of future mobility, the fourth workshop was conducted with twelve project stakeholders to co-design these visions further through an iterated design canvas created by the researchers. In the fifth and last workshop, we invited two key stakeholders and two community champions from Biskopsgården. The workshop aimed at co-designing and iterating on the two mobility ideas from the previous workshop. The focus was to evaluate, develop, and anchor the design ideas in relation to the socio-physical context of Biskopsgården through a concept summary page and a geographical map of the area.

Activity	Aim	Purpose	Type of participant(s)	Nr. Of participants
Workshop 1	Exploring and ideation	Create a common ground	Project stakeholders	14
Workshop 2	Exploring and ideation	Understanding of current routines and potential futures	Inhabitants of Biskopsgården	26
Workshop 3	Exploring and ideation	Understanding of current routines and potential futures	Intending buyers of Forsåker apartments	2
Participatory Observation 1	Exploring	Gaining insight on the social fabrics of context	Community champion, Biskopsgården	1
Participatory Observation 2	Exploring	Insight on a car-free, multimodal lifestyle	Community champion nearly located to Forsåker	1
Workshop 4	Co-designing	Responsibilities and functionalities in future mobility hubs	Project stakeholders	12
Workshop 5	Co-designing	Anchoring and tailoring to a socio-physical context	Two key stakeholders and two community champions	4

Table 1. Research activity aims and participants.

3.1. Analysis

The research was iterative and inductive, shaped by interpretivist and phenomenological traditions that emphasise participants' lived experiences and the meanings they attributed to the design process. As O'Reilly (2012) highlights, detailed fieldnotes were central in capturing not only activities, but also relational dynamics, atmospheres, and emergent themes. These notes, alongside workshop artefacts and post-session reflections, were analysed continuously concerning project developments and relevant literature. Rather than testing fixed hypotheses, our approach allowed insights to develop over time through observation, reflection, and reinterpretation cycles. This analytical process of "constant comparison" (described in relation to a) ethnography in O'Reilly (2012) and b) design ethnography in information systems in Baskerville & Myers (2015)) helped us identify patterns in how participants engaged with each other, interpreted mobility in daily life, and co-develop locally grounded design proposals. For example, participants felt purposefully left out of the rest of the city due to a lack of mobility options and long tram routes, even though areas of interest were geographically close, leading to digital chat groups being formed for sharing monthly transport cards and everyday material.

Design ethnography builds on the ethnographic commitment to understanding practices in situ, but aligns this with a future-oriented, speculative agenda characteristic of design (Pink et al., 2022). It acknowledges that "the field" is not a neutral backdrop, but actively shaped through methods, relationships, and the unfolding design inquiry. In keeping with these analytical principles of design ethnography, the results in the next section are not reported as isolated findings but as emergent insights shaped through iterative engagements with diverse participants across multiple contexts. Each workshop and observational encounter were a site of inquiry and a generative moment within a broader co-design process. Accordingly, the structure of the results follows the chronological unfolding of fieldwork, emphasising how local knowledge, social relations, and spatial dynamics informed the development of thematic categories and design ideas. This structuring reflects the epistemological stance that insights are not merely discovered but co-produced through interaction, negotiation, and participation (Donovan & Gunn, 2016). This approach reflects our commitment to generating contextually grounded and collaboratively produced knowledge - hallmarks of ethnographically informed design research within Information Systems (Blomberg & Karasti, 2013).

4. Findings: from the field to design ideas for the future

This section presents each workshop and participant observation as a source of insight and a transformative moment that shaped subsequent activities. This chronological structure reveals how learning accumulated across sites and actors, leading to iterative refinement of thematic understanding and design propositions. This approach serves two purposes. First, it provides transparency into the processual nature of knowledge production, allowing readers to trace how specific insights gained relevance in context. Second, it reflects the methodological imperative of participatory and co-design research to foreground the voices, values, and practices of those involved (Ehn et al., 2014; Fors et al., 2025). The results remain grounded in context by situating each phase of engagement within the lived realities of communities in Biskopsgården and Forsåker. The section culminates in synthesising two overlapping but distinct themes: one emerging from stakeholder perspectives and one from residents' lived experiences. These thematic clusters informed the development of two design ideas that aimed to respond to local values and social practices surrounding mobility in everyday life. In line with the design ethnographic tradition, these ideas are not presented as fixed solutions, but as situated responses to specific spatial, cultural, and social conditions - prototypes for further dialogue, rather than endpoints (Pink et al., 2018). What follows is a chronological account of these engagements, interlinked with reflexive accounts of learnings from each iteration, foregrounding the situated nature of the findings and their contribution to the co-design of locally grounded mobility futures.

4.1. Setting the Stage

The first workshop aimed to create a shared understanding of what a mobility hub could and should be and started with an introduction to the phenomenon of future mobility and state-of-the-art research on the topic. The workshop focused on identifying pre-existing social values that could be emphasised in future mobility hubs and explored how the partners' resources could shape such hubs' potential content and functionalities. The ideas and resources available were discussed in groups of six to seven participants, documented on a whiteboard, and later digitised. Researchers later used workshop participants' vision(s) to communicate and provoke reactions among residents in Biskopsgården and Forsåker.

Learnings from the session highlighted the complexity of building a shared vocabulary and

direction across organisational boundaries. Participants came from diverse backgrounds with different understandings of mobility shaped by their specific experiences and professional goals. As a result, what "future mobility" meant varied widely as some gravitated toward active transport solutions like cycling, while others were focused on questions of market viability and operational models. This led to repeated efforts to introduce and maintain a focus on social sustainability and aspects of human flourishing. The need to foreground social values was not immediately self-evident to all stakeholders. Thereby, it became important to align future visions and make space for negotiations about the terms through which such visions are framed.

4.2. Approaching the community

Workshop two was conducted with residents of Biskopsgården. This workshop focused on current mobility routines and generated ideas for mobility hub services based on participants' goals and everyday needs. The sessions started with an introduction to the research project and the aim of designing alternatives to future mobility services and systems. Afterwards, groups of four to five individuals were handed a paper on which they were to illustrate their everyday mobility routines together with an empirical data management contract. A problem regarding trust surfaced as the institutional language of such contracts seemed to raise scepticism towards the research group, but after 20-30 minutes, the concern was solved together with the community champion, and the activity proceeded. After 30 minutes of visualisation, a scenario was presented, including a private car ban substituted by a mobility hub with associated services. However, it is worth noting that only one out of the 26 participants owned a car. Insights from the workshop indicated that many participants felt disconnected from the rest of the city due to a lack of micromobility alternatives and long travel distances to geographically close destinations. Several ideas revolved around sharing everyday items, such as toys, books, and reused furniture. Participants expressed strong interest in continued involvement, and key individuals were identified to support further citizen recruitment and engagement in the project. We came to call these key individuals "champions".

This workshop served as a moment of personal and professional confrontation with the limitations of our assumptions as researchers. Despite arriving with participatory tools and a co-design agenda, we realised how academic, institutional habits attempted to shape the engagement structure but failed. It became evident that what we had initially considered as "neutral" workshop techniques, with fixed timeslot, formalised worksheets, and scenario-based prompts, reflected a perspective grounded in privileged research positions rather than locally lived experiences. Many participants had long histories of unfulfilled promises and extractive projects. Our understanding of "community-based design" was overly influenced by project procedural aspects rather than contextual tuning. This underscored the need to re-evaluate the former design perspective and to remain responsive to the social and historical conditions shaping community participation. Community-based design is thereby not about inviting others into a predefined process, but rather to co-create the structure and rhythm of that process together.

4.3. Exploring the future

The third workshop was an online activity and included two future residents of Forsåker, serving both as a pilot for a scalable online format (since no one currently lived in the area) and to gain deeper insight into mobility routines and "must-haves" for future mobility hubs. Once again, researchers described the project's aims and goals before moving into the active part of the session. Participants were asked to highlight current routines experienced as positive, including both car- and non-car-related practices. In the latter part of the activity, participants were asked to describe what was unacceptable in a potential future with mobility hubs as substitutes for privately owned cars. Notes, including the topics brought up by participants, were co-created in Miro.

Insights from the third workshop emphasised that mobility often extends beyond the individual and must accommodate responsibilities toward family members, visitors, and friends. Participants also highlighted current challenges with transitioning to sustainable transport, noting how car-centric priorities persist in municipal and regional planning. For example, infrastructure for sustainable modes was reportedly deprioritised during winter compared to car infrastructure. Furthermore, the session showed us that community-driven design is not solely tied to existing communities but can also serve as a tool for imagining and co-creating future ones. Even without local dynamics, it is possible to reflect on what kind of social and infrastructural conditions people hope to see.

4.4. Anchoring voices

The first participatory observation was conducted in Biskopsgården with a local community champion, a local stakeholder with strong insights into community values and practices. Throughout the day, we took part in the everyday life of this individual, who took part in meetings regarding initiatives, enabling access to youth

activities, and facilitated a monthly meeting with community members ranging from citizens and associations to politicians and local business owners. The first session deepened the relationship established during workshop two and aimed to develop an understanding of communication norms and local engagement practices. The observation revealed how information is disseminated through speaker lists at local events and provided insight into perceptions and usage of ongoing mobility initiatives in the area. Everyone who wanted to say something could sign up on the speaker list, and every speaker got one minute each. The activity also facilitated contact with local associations, creating opportunities for future collaboration, primarily with social organisations needing support for their mobility routines.

The second participant observation occurred with a community champion envisioning future residence in Forsåker and built upon the themes from workshop 3. We followed the champion, who had transitioned from using a privately owned car to utilising a multimodal, partially owned mobility system. The session brought us to different city areas, from the centre to the outskirts, combining parental duties of baby care with initiatives for local slow-moving vehicle drivers (youngsters driving EPAs) and work-related meetings.

The participatory observations enabled us to observe and participate in the flourishing aspects of everyday life, extending the vision of pure work-related efficiency to surrounding community perspectives. They provided a deeper understanding of the practical and social challenges of living a car-free, multimodal life. They also highlighted the knowledge and social skills required to maintain such a lifestyle, particularly regarding informal lending systems and shared access to services. Furthermore, the observations highlighted how champions acted as mediators between institutional frameworks and community perspectives by anchoring abstract mobility ideas within meaningful social contexts. Their established trust within broad local networks enabled conversations that allowed co-shaping of the process, underscoring the importance of recognising and supporting such intermediaries in community-driven design. Furthermore, the observations showed glimpses of champions' power in distributing resources and enabling access for others – be it through guidance of where to turn for assistance in private life, joining a chat group of specific themes, or borrowing physical keys for unlocking doors.

Following the three workshops and two observations, we compiled a summary of insights, which became the basis for creating friction cards. These friction cards were employed in a subsequent workshop to challenge prevailing assumptions and stimulate design thinking. Each friction card combined a quotation from a participant in earlier research activities, a summary of a persistent issue related to or influenced by mobility, and a prompting question intended to spark discussions and ideation around future mobility hubs. In this way, the friction cards functioned as a tool for anchoring voices: they ensured that the perspectives, experiences, and concerns of community members remained central to the co-design process, actively shaping both the evaluation of existing ideas and the generation of new ones.

4.5. Co-designing ideas

The fourth workshop focused on communicating and iterating the emerging design ideas with project stakeholders through a lens of locally grounded values. The session aimed to concretise design ideas while addressing questions and receiving critical feedback from project stakeholders. Two canvas tools created by the researchers were used to flesh out the ideas for the two main ideas – Hyperlocal Mobility and Community-based Mobility - while the friction cards stimulated discussion and reflection.

The fifth and final workshop involved key stakeholders, local champions, and project partners to refine the design ideas further collaboratively, together with a research team member. For this session, maps of the two areas were used to ground the design ideas in actual places, along with summary sheets outlining associated service ideas that could assist researchers, stakeholders, and champions in visualising potential community-based mobility solutions. The main aim of the co-design workshops was to discuss project findings and make sense of these through active discussion regarding design ideas. The workshop opened for actors who do not usually meet, enabling a direct flow of local knowledge and area expertise between community champions and project stakeholders. While these groups did not always align regarding priorities or perspectives, the setting provided a structured opportunity for constructive dialogue.

4.6. Future mobility ideas

This study identified two sets of themes, one from stakeholders and one from residents, that informed two design ideas focused on local values and everyday life. Stakeholders emphasised future mobility hubs with themes of durability (integration with public transport and long-term adaptability), community-building (supporting local associations and participation), independence (enabling car-free living), and connectivity (embedding hubs in daily social and spatial contexts). Residents, drawing from lived mobility experiences, highlighted inclusion (exclusion caused by

transport gaps, especially for caregivers and people with disabilities), the importance of local infrastructures (trust-based networks led by community champions), a preference for human presence over surveillance for safety, flexible and collective use (services supporting household and shared access), and the need to rethink territorial boundaries to reflect social geographies rather than administrative ones. These insights shaped two socially attuned design ideas:

Hyperlocal Mobility seeks to harness the capacity of neighbourhood-based networks and everyday social relations to create flexible solutions that address specific gaps in provision. By foregrounding practices of care, mutual support, and local knowledge, it emphasises mobility as movement through space and participation in community life. Crucially, it proposes a mobility network that circulates within the geographical area itself, rather than functioning solely to transport people in and out of the area towards the urban centre. In doing so, it strengthens local ties and supports the everyday infrastructures that sustain social belonging.

Community-Based Mobility, in turn, underscores the institutional potential of established associations and civic groups to act as mediators between residents and formal transport systems. It builds on existing communities and their established ways of organising mobility, positioning these practices as the foundation for a model in which business and service provision turn to the community as a collective actor rather than addressing individuals in isolation. This design idea extends beyond service delivery to encourage co-ownership and collective responsibility, embedding mobility within the shared practices that sustain social resilience.

5. Discussion

This paper explored how a community-driven design approach can be used to address social sustainability and human flourishing in mobility systems. Grounded in design ethnography and co-design workshops, the process facilitated fertile ground for local perspectives, challenging dominant assumptions and allowing local values and lived experiences to shape mobility futures, reflecting everyday tensions and aspirations over time. As sustainable design projects tend to neglect social aspects historically, such processes are highly welcomed (Corsini & Moultrie, 2021). In our case, co-design methods enabled the framing of "microcosms" (Mazé et al., 2011) of everyday mobility and social relations, wherein residents articulated their visions for mobility solutions that reflect collective needs, values, and aspirations. These situated design practices are referred to by Jones (2017) as "designing for flourishing cultures", where local community networks, rather than top-down systems, serve as the organising logic for innovation. Future research could explore these flourishing cultures further, especially in understanding how they could be supported long-term, even after project deadlines (Corsini & Moultrie, 2021).

The research activities established a ground for relationships between citizens and stakeholders that opened a participatory milieu for exploring socially sustainable, locally grounded alternatives to current norms. As such, the activities aimed to create engagements for differing perspectives for a compassionate environment (Ciriello, 2025). This way of working proved valuable in exposing the importance of rethinking the temporal and spatial logic underpinning traditional planning and designing. For example, neighbourhood boundaries used in official planning often failed to correspond to actual social geographies, enhancing the feeling of intentional exclusion of inhabitants. The process made this disconnect visible, enabling design responses that better reflect how people live and move. The approach is an example of how flourishing can be operationalised through design, and thereby a step towards how flourishing can be defined in practice, as called for by Corsini & Moultrie (2021). From this understanding of disconnection, the design ideas emerged. The design ideas are not endpoints, rather concretisations of community logic shaped by the iterative interplay between method, material, and lived community experiences. Such iterative interplay could assist in futuristic speculation of solutions and simultaneously co-design and adjust methods that are locally attuned to community values.

The design ideas illustrate the type of insights that become possible when design research focuses not only on problem-solving but also on the socially integrative spaces that blend digital and physical infrastructure. As such, the design ideas function as a catalyst for participation and inclusion during the research process. This echoes Ferdman's (2024) notion of technology as an affordance that must be aligned with human strengths and communal wellbeing. For example, by redefining business models, the study highlights how design can facilitate new governance structures rooted in community values. More broadly, this process highlights a shift in the role of designers as they become facilitators rather than service providers. In line with Jones (2017) and Pohlmeyer (2017), we argue that designing for flourishing involves creating platforms where communities can co-develop the conditions for autonomy and meaning. However, such processes are fragile, as a lack of long-term commitment may risk reinforcing problems regarding exclusion and gentrification (Eizenberg & Jabareen, 2017). Relating to

Corsini & Moultrie's (2021) future research agenda, our approach aimed to design solutions that preserved the communities' social systems and shed light on fundamental obstacles for these systems. However, as the work is on a conceptual level, no evaluation of the design ideas was conducted, meaning further work is needed to explore the potential impact of the design ideas on human flourishing.

Another key contribution is articulating mobility as a service and a form of social infrastructure. In line with Jones (2017), framing flourishing as relational, and as an attempt to operationalise human flourishing (Corsini & Moultrie, 2021), we argue that mobility should be understood as a capability shaped by informal infrastructures and local practices. In mobility, integrating community champions, community-based access models, and hyperlocal reuse economies in our design interventions, speaks directly to this type of framing. These insights are arguably transferable to other empirical fields, especially as they relate to calls for understanding history, traditions, and dialogue – central aspects to local values and human flourishing (Corsini & Moultrie, 2021). Finally, we call for greater attention to design that enables shared ownership. This includes participatory mechanisms and service flexibility that allow communities to shape and adapt systems over time. The notion of alternative, fair community-based business models serves as a conceptual provocation and service provision where local organisations take on more active roles and responsibilities in shaping their environments.

Overall, this study shows the potential of community-centred, ethnographically grounded design as a meaningful tool for reshaping mobility systems that reflect and reinforce the local cultural ecosystems necessary for human flourishing. It provides an example of a digitally supported design honouring lived experiences as an alternative to current efficiency and capital profit norms (Ciriello, 2025). However, it is worth noting that good intentions may have unintended consequences, especially as experiences are dynamic and situational, thereby changing over time (Ciriello, 2025). Therefore, studies that commit to extended periods are needed (Corsini & Moultrie, 2021), where continuous evaluation and iteration of roles, service functionalities, and power dynamics are studied. Additionally, considering mobility, it becomes increasingly important for designers to be aware of long-term agendas and plans to make active choices regarding avoiding the fuelling of gentrification, especially when processes are project-oriented or timebound (Eizenberg & Jabareen, 2017). Hence, it is naive to rely on a specific approach or method to guarantee human flourishing, which implies further complications regarding how flourishing can be understood and operationalised. Furthermore, the community-driven design process may function as a continuously evolving understanding of local values associated with flourishing instead of a conceptual, static "agreement", while also – in line with Ciriello (2025) - being an alternative space where diverse experiences can support genuine connections over profit-oriented agendas. Importantly, our study revealed that these processes were already present in the research contexts, particularly in Biskopsgården, highlighting that practices oriented toward wellbeing and community cohesion often exist organically. As calls for design research increasingly emphasise moving away from harm avoidance toward proactively enabling wellbeing (Jones, 2017; Pohlmeyer, 2017; Corsini & Moultrie, 2021), our findings point to the potential of community-driven processes to shape mobility in ways that foreground wellbeing and local flourishing through the design ideas culminating from the outlined design approach. Hyperlocal Mobility emphasises informal, place-bound, and bottom-up practices that circulate everyday social life within the neighbourhood, and Community-Based Mobility highlights more organised, collective approaches through local associations. Together, they illustrate complementary pathways grounded in lived experience, local exchange, and institutionalised shared responsibility, demonstrating how mobility systems can be efficient and socially embedded. These insights indicate fruitful directions for future research to explore how such approaches can be identified, supported, and scaled, leveraging local resources and everyday practices to promote equitable and resilient mobility.

5.1. Future research

Building on the design approach outlined in this article and the design ideas of Hyperlocal and Community-Based Mobility culminating from this process, future research could examine the practical mechanisms through which these ideas can be integrated into urban planning and transport policy, particularly in areas affected by transport poverty. There is a need to evaluate how community-driven mobility models influence social cohesion, wellbeing, and resilience over time, and to identify the conditions under which such approaches are most effective. Additional research could investigate the scalability of these concepts across different urban contexts and how local resources, social networks, and everyday practices can be systematically leveraged to support inclusive and sustainable mobility systems. Furthermore, this research could contribute to a deeper understanding of how the digitalisation of mobility services might serve as a provider of mobility justice, rather than reinforcing existing inequalities

(Sheller 2018). Finally, exploring the interactions between bottom-up community initiatives and formal institutional frameworks would provide valuable insights into how design for human flourishing can be operationalised at multiple levels.

6. Conclusion

This study demonstrates how a community-driven design approach can address human flourishing in mobility by embedding design processes in local values, lived experiences, and collective aspirations. Through co-design and ethnographic engagement, alternative logics of infrastructure and governance emerged, highlighting how mobility can be reframed as a relational capability and form of social infrastructure rather than a purely technical service. However, the findings underscore the need for long-term, iterative processes that continuously adapt to community dynamics and guard against risks of exclusion or gentrification. In conclusion, a community-driven design has the potential to enhance the functionality of systems and embed them within the social and cultural fabric of communities. By centring on lived experiences and mobilising local organisational capacities, such an approach holds promise for designing more flourishing, equitable, inclusive, and resilient futures.

7. References

Baskerville, R. L., & Myers, M. D. (2015). Design ethnography in information systems. Information Systems Journal, 25(1), 23-46.

Blomberg, J., & Karasti, H. (2013). Reflections on 25 years of ethnography in CSCW. Computer supported cooperative work (CSCW), 22, 373-423.

Boyer, R. H., Peterson, N. D., Arora, P., & Caldwell, K. (2016). Five approaches to social sustainability and an integrated way forward. Sustainability, 8(9), 878.

Ciriello, R. F. (2025). Dear schopenhauer: How can is artefacts help us flourish? Wille, Vorstellung, compassion, and beauty in social media. Scandinavian Journal of Information Systems, 37(1), 9.

Corsini, L., & Moultrie, J. (2021). What is design for social sustainability? A systematic literature review for designers of product-service systems. Sustainability, 13(11), 5963.

Dillahunt, T. R., & Veinot, T. C. (2018). Getting there: Barriers and facilitators to transportation access in underserved communities. ACM Transactions on Computer-Human Interaction (TOCHI), 25(5), 1-39.

Donovan, J., & Gunn, W. (2016). Design anthropology: An introduction. Design and Anthropology, 1-16.

Ebbesson, E., Lund, J., & Smith, R. C. (2024). Dynamics of sustained co-design in Urban Living Labs. CoDesign, 20(3), 422-439.

Ehn, P., Nilsson, E. M., & Topgaard, R. (2014). Making futures: Marginal notes on innovation, design, and democracy (p. 392). The MIT Press.

Eizenberg, E., & Jabareen, Y. (2017). Social sustainability: A new conceptual framework. Sustainability, 9(1), 68.

Ferdman, A. (2024). Human flourishing and technology affordances. Philosophy & Technology, 37(1), 1.

Fors, V., Brodersen, M., Raats, K., Pink, S., & Smith, R. C. (2022). Investigating ADM in Shared Mobility: A design ethnographic approach. In Everyday Automation (pp. 197-212). Routledge.

Fors, V., Smith, R.C., Pink, S., Lund, J. & Ebbesson, E. (2025). Disrupting intelligent mobilities systems: Design ethnographic pedagogy for multi-stakeholder innovation. Ethnographic Praxis in Industry Conference Proceedings EPIC ‘25 (OA)

Jones, P. (2017). Social ecologies of flourishing: Designing conditions that sustain culture. In Design for a Sustainable Culture (pp. 38-54). Routledge.

Martens, K. (2016). Transport justice: Designing fair transportation systems. Routledge.

Mazé, R., Gregory, J., & Redström, J. (2011). Social Sustainability: A design research approach to sustainable development.

Mejía Dorantes, L., & Murauskaite-Bull, I. (2022). Transport Poverty: A systematic literature review in Europe (No. EUR 31267 EN).

Miskolczi, M., Földes, D., Munkácsy, A., & Jászberényi, M. (2021). Urban mobility scenarios until the 2030s. Sustainable Cities and Society, 72, 103029.

Nakata, C. (2025). Designing innovations for and with human flourishing: A research agenda. The Design Journal, 28(1), 50-53.

O'reilly, K. (2012). Ethnographic methods. Routledge.

Pink, S., Akama, Y., & Sumartojo, S. (2018). Uncertainty and possibility: New approaches to future making in design anthropology. Bloomsbury Publishing.

Pink, S., Fors, V., Lanzeni, D., Duque, M., Sumartojo, S., & Strengers, Y. (2022). Design ethnography: Research, responsibilities, and futures. Routledge.

Pohlmeyer, A. E. (2017). How design can (not) support human flourishing. Positive psychology interventions in practice, 235-255.

Rossi, E. (2025). A design culture on social inclusion and sustainability. Innovation Design for Social Inclusion and Sustainability: Design Cultures and Creative Practices for Urban Natural Heritage, 17.

Sheller, M. (2018). Mobility justice: The politics of movement in an age of extremes. Verso Books.

Smith, R. C., Fors, V., Brodersen, M., Lund, J., & Ebbesson, E. (2024). Sustainable Automated Futures: Participatory Human Approaches to Urban Mobility. The De Gruyter Handbook of Automated Futures, 413.

Weberg, O., Fors, V., & Lund, J. (2025). Scaling Deep with Local Community Champions in Living Labs. Sustainability, 17(13), 5888.